

# PersonaPulse AI

## Google Ads API Tool Design Documentation

Prepared for Google Ads API Developer Token Review

Company: Cobweb Arquitetura e Consultoria Dados

CNPJ: 67.083.842/0001-25

Contact email: cloud.datascience.arch@gmail.com

Application URL: <https://personapulse-ai.onrender.com/app>

API health check: <https://personapulse-ai.onrender.com/health>

Public application URL: <https://personapulse-ai.onrender.com/app>

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Purpose: This document describes the PersonaPulse AI tool architecture, OAuth flow, data collection scope, user interface, permissions, security controls, and Google Ads API usage. It intentionally excludes confidential credentials, secrets, tokens, and customer private data.

## 1. Executive Summary

PersonaPulse AI is a marketing intelligence and campaign performance platform operated by Cobweb Arquitetura e Consultoria Dados. The tool helps authorized business users consolidate first-party customer records, campaign metrics, product price intelligence, marketing recommendations, and executive dashboards in one clean interface.

For Google Ads, PersonaPulse AI uses the official Google Ads API only after OAuth authorization by an authorized account user. The integration is designed to read campaign and performance metrics, normalize the data into PostgreSQL, and display campaign-level and executive-level results in PersonaPulse AI and Power BI dashboards.

The tool does not scrape Google Ads, does not bypass account permissions, does not collect Google passwords, does not collect end-user advertising identifiers, and does not resell raw Google Ads data. Google Ads data is used only to support the authorized business account owner with campaign measurement and optimization.

## 2. Product Scope

- Provide an executive dashboard for clients, revenue, campaign performance, ROI, ROAS, consent status, and recommended actions.
- Receive consented CRM and CSV data such as customer records, orders, and behavioral events.
- Connect advertising APIs, beginning with Google Ads, to retrieve campaign performance for authorized accounts.
- Generate editable campaign briefs and ad copy suggestions when a user explicitly chooses to use AI assistance.
- Provide structured Power BI endpoints for business reporting.

- Support price research using structured data providers such as DataForSEO Merchant API, separate from the Google Ads integration.

### 3. High-Level Architecture

PersonaPulse AI is deployed as a web application and API service. The current MVP architecture is intentionally simple so that security, data lineage, and API usage can be audited clearly.

| Layer         | Component                                                 | Role                                                                                                                            |
|---------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Frontend      | HTML, CSS, JavaScript single-page application             | Dashboard, client list, campaign generator, integrations, LGPD, price research, active campaigns.                               |
| Backend API   | Python API service hosted on Render                       | OAuth handling, CRM endpoints, Google Ads synchronization, Power BI endpoints, data normalization.                              |
| Database      | PostgreSQL on Render                                      | Persistent storage for customers, orders, events, campaigns, campaign metrics, recommendations, price research, and audit logs. |
| External APIs | Google Ads API, CRM integrations, DataForSEO Merchant API | Authorized campaign metrics, business data ingestion, product price intelligence.                                               |
| BI Layer      | Power BI JSON endpoints                                   | Executive reporting using aggregated, normalized data.                                                                          |

### 4. System Data Flow

1. A business user opens PersonaPulse AI through the HTTPS web application URL.
2. The user imports first-party data from CSV or synchronizes an authorized CRM source.
3. The user configures the Google Ads connector in the Integrations screen, using an OAuth client, developer token, customer ID, and optional manager account ID when applicable.
4. The user starts OAuth authorization. PersonaPulse AI redirects the browser to Google, and Google returns an authorization code to the configured callback endpoint.
5. The backend exchanges the authorization code for tokens through Google OAuth, stores authorization state server-side, and uses the access token to call the Google Ads API.
6. The backend retrieves campaign-level metrics, normalizes them, writes them to PostgreSQL, and exposes them to the dashboard and Power BI endpoints.

### 5. Google OAuth Flow

PersonaPulse AI uses OAuth to ensure that Google Ads data is accessed only after a user with the correct account permissions authorizes the application. The OAuth callback currently used by the production service is:

<https://personapulse-ai.onrender.com/api/oauth/callback>

| Step | Actor | Description |
|------|-------|-------------|
|------|-------|-------------|

|   |                 |                                                                                                                                          |
|---|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | User            | Clicks Open OAuth in PersonaPulse AI after configuring the Google Ads connector.                                                         |
| 2 | PersonaPulse AI | Generates a state value and redirects the browser to Google OAuth with the Google Ads scope.                                             |
| 3 | Google          | Displays the consent screen and validates the Google account authorization.                                                              |
| 4 | Google          | Redirects back to the PersonaPulse AI callback URL with authorization code and state.                                                    |
| 5 | PersonaPulse AI | Validates state, exchanges code for tokens, and stores authorization data server-side.                                                   |
| 6 | PersonaPulse AI | Calls Google Ads API endpoints only for the authorized customer account or approved client accounts under an authorized manager account. |

## 6. Google Ads API Permissions

The Google Ads integration requests the Google Ads API OAuth scope:

<https://www.googleapis.com/auth/adwords>

This scope is required to retrieve Google Ads account and campaign performance data through the official Google Ads API. PersonaPulse AI uses this permission for authorized campaign reporting and performance analysis only. It does not use the permission to access unrelated Google products, private Gmail data, Drive files, passwords, or personal browsing data.

## 7. Google Ads Data Collected

The Google Ads connector collects campaign and performance data necessary to calculate business metrics and display campaign dashboards. The collected fields may include:

- Google Ads customer ID and optional manager account context.
- Campaign ID, campaign name, campaign status, and advertising channel type.
- Metric date or reporting period when included in the query.
- Impressions, clicks, conversions, conversion value or attributed revenue, and cost.
- Derived metrics calculated by PersonaPulse AI, including CTR, conversion rate, ROAS, ROI, cost over revenue, and net result.

PersonaPulse AI does not collect Google account passwords, payment card information, Google user cookies, raw end-user identifiers, or personally identifiable Google Ads audience membership data.

## 8. Non-Google Data Collected by PersonaPulse AI

PersonaPulse AI may also ingest non-Google business data when authorized by the business user. This data is used to connect campaign performance with business outcomes.

| Source | Example Data | Purpose |
|--------|--------------|---------|
|--------|--------------|---------|

|                         |                                                                             |                                                    |
|-------------------------|-----------------------------------------------------------------------------|----------------------------------------------------|
| CRM or CSV              | Customer name, email, phone, city, consent flag, source, purchase behavior. | Customer segmentation and campaign eligibility.    |
| Orders                  | Product, category, store, order value, purchase date.                       | Revenue attribution and customer profile analysis. |
| Events                  | Lead activity, cart abandonment, interest signals, support or sales events. | Recommendation and segment scoring.                |
| DataForSEO Merchant API | Structured product prices from permitted commerce sources.                  | Market price research and price suggestion.        |
| Power BI                | Aggregated dashboard snapshots.                                             | Executive reporting and visual analytics.          |

## 9. User Interface Screens

The PersonaPulse AI interface is organized by operational workflow. The following screens are available in the MVP:

- **Executive Dashboard:** consolidated clients, revenue, campaigns, consent, recommendations, and campaign performance.
- **Clients:** imported or synchronized customer records, including origin, product, value, purchase channel, and consent status.
- **Segments:** behavioral segments and intent scores created from authorized business data.
- **Campaign Generator:** editable campaign brief fields, with optional AI generation only when the user explicitly requests it.
- **Active Campaigns:** campaign activation status and individual performance dashboard for selected campaigns.
- **Price Research:** product pricing references using structured commerce data providers.
- **Integrations:** API connector configuration for CRM, Google Ads, other Ads platforms, OAuth, and credentials.
- **Recommendations:** daily improvement recommendations for campaign and data quality.
- **LGPD:** consent and privacy-oriented controls for marketing eligibility.

## 10. API Endpoints Used by Integrations and BI

| Endpoint             | Method   | Purpose                                                                            |
|----------------------|----------|------------------------------------------------------------------------------------|
| /health              | GET      | Service health, persistence status, database connection state.                     |
| /api/oauth/start     | GET      | Starts OAuth flow for supported providers such as Google Ads.                      |
| /api/oauth/callback  | GET      | Receives OAuth authorization response and exchanges authorization code for tokens. |
| /api/connectors/save | POST     | Stores connector configuration server-side.                                        |
| /api/connectors/sync | POST     | Runs authorized API synchronization, including Google Ads.                         |
| /api/crm/customers   | GET/POST | Receives and lists CRM customer data.                                              |
| /api/crm/orders      | GET/POST | Receives and lists CRM order data.                                                 |
| /api/crm/events      | GET/POST | Receives and lists CRM behavior events.                                            |

|                                |      |                                                                   |
|--------------------------------|------|-------------------------------------------------------------------|
| /api/campaigns/generate        | POST | Generates or prepares campaign briefs when requested by the user. |
| /api/powerbi/executive-summary | GET  | Executive summary JSON for Power BI.                              |
| /api/powerbi/customers         | GET  | Customer rows for Power BI.                                       |
| /api/powerbi/campaigns         | GET  | Campaign rows and metrics for Power BI.                           |
| /api/powerbi/sources           | GET  | Data source status rows for Power BI.                             |

## 11. PostgreSQL Storage Model

PersonaPulse AI persists operational data in PostgreSQL when DATABASE\_URL is configured in the production environment. The storage model is divided into schemas for application data, integrations, business intelligence snapshots, and audit logs.

- app.customers, app.orders, app.events: first-party customer and transaction data.
- app.campaigns, app.campaign\_metrics: campaign definitions and performance metrics from Ads sources.
- app.price\_researches: structured product price research results.
- app.recommendations: optimization recommendations and campaign suggestions.
- integrations.data\_sources and integrations.connector\_configs: connector metadata and configuration state.
- bi.powerbi\_snapshots: consolidated reporting snapshots for Power BI.
- audit.audit\_logs: audit trail for data sync, connector actions, OAuth results, and relevant administrative actions.

## 12. Security Controls

- All production traffic uses HTTPS through Render hosted URLs.
- OAuth redirect URI uses HTTPS and a server-side callback endpoint.
- OAuth state is generated and checked to reduce cross-site request forgery risk in the authorization flow.
- Client secrets, refresh tokens, developer tokens, database URLs, and third-party API passwords are not committed to GitHub and are stored as environment variables or server-side configuration.
- The browser frontend does not need direct access to Google client secrets or developer tokens in the production design.
- PostgreSQL is the production persistence layer, with application tables separated by schema and with audit logging for critical actions.
- PersonaPulse AI applies data minimization: it requests only fields needed for dashboarding, performance calculation, reporting, and campaign optimization.
- Users can separate data sources such as CSV, CRM, Meta Ads, Google Ads, and other Ads sources so that source lineage is visible in the dashboard.
- LGPD-oriented consent flags are maintained, and non-consented customers are excluded from marketing recommendations where applicable.

## 13. Google Ads API Usage Policy Alignment

PersonaPulse AI is designed to use the Google Ads API in a limited, transparent, and user-authorized way. The integration is used for reporting and optimization of the advertiser accounts that authorize the application.

- No unauthorized access: the app accesses only Google Ads accounts authorized through OAuth and account permissions.
- No scraping: the app uses the official Google Ads API, not browser scraping or automated UI extraction.
- No resale of raw data: Google Ads data is used to provide reporting and optimization to the authorized business user.
- No secret exposure: developer tokens, client secrets, OAuth refresh tokens, and database credentials are not exposed in public frontend code.
- No sensitive end-user profiling from Google Ads: the integration focuses on account and campaign performance metrics.
- No automated spend changes in the MVP: the MVP is focused on reading metrics and supporting decisions. Campaign publishing or budget changes, if added later, will require explicit user action and separate permission review.

## 14. Current MVP Status and Access Request Reason

The PersonaPulse AI MVP has working CRM synchronization, PostgreSQL persistence, Power BI endpoints, campaign dashboards, and the Google Ads connector configuration screen. The current blocker for production Google Ads synchronization is Google Ads API access level. The developer token currently returns a `DEVELOPER_TOKEN_NOT_APPROVED` error for non-test accounts, which indicates that Basic or Standard access is required before the tool can retrieve real production account metrics.

This document supports the access review by explaining the tool design, intended API usage, data handling, and security model.

## 15. Reviewer Test Path

7. Open the PersonaPulse AI application at <https://personapulse-ai.onrender.com/app>.
8. Open the Integrations screen and choose Google Ads.
9. Confirm that the connector requests Google Ads OAuth access for authorized campaign metrics.
10. Authorize an approved Google account with access to the intended Google Ads customer account.
11. Run Sync Real to retrieve campaign metrics through the Google Ads API.
12. Review the Executive Dashboard, Active Campaigns screen, and Power BI endpoints to confirm normalized metrics and source separation.